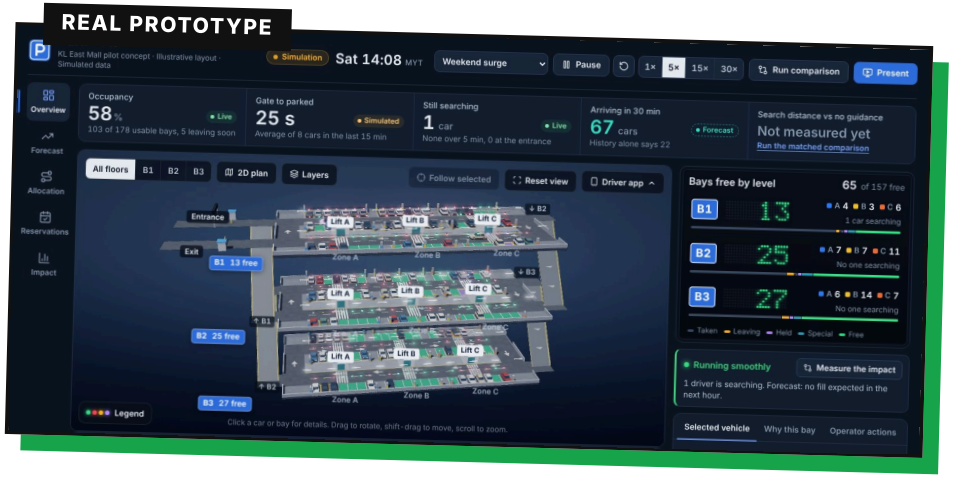
PARKFLOW

RESERVE CAPACITY, NOT PARKING BAYS.

A bookable **Priority Zone** by the lifts, free for a mall's best customers, shared with everyone else, and a forecast that tells every driver where to park.

PILOT: KL EAST MALL **TEAM HEXA**



Prototype screen uses simulated data on an illustrative layout.

TEAM · 00

TEAM HEXA

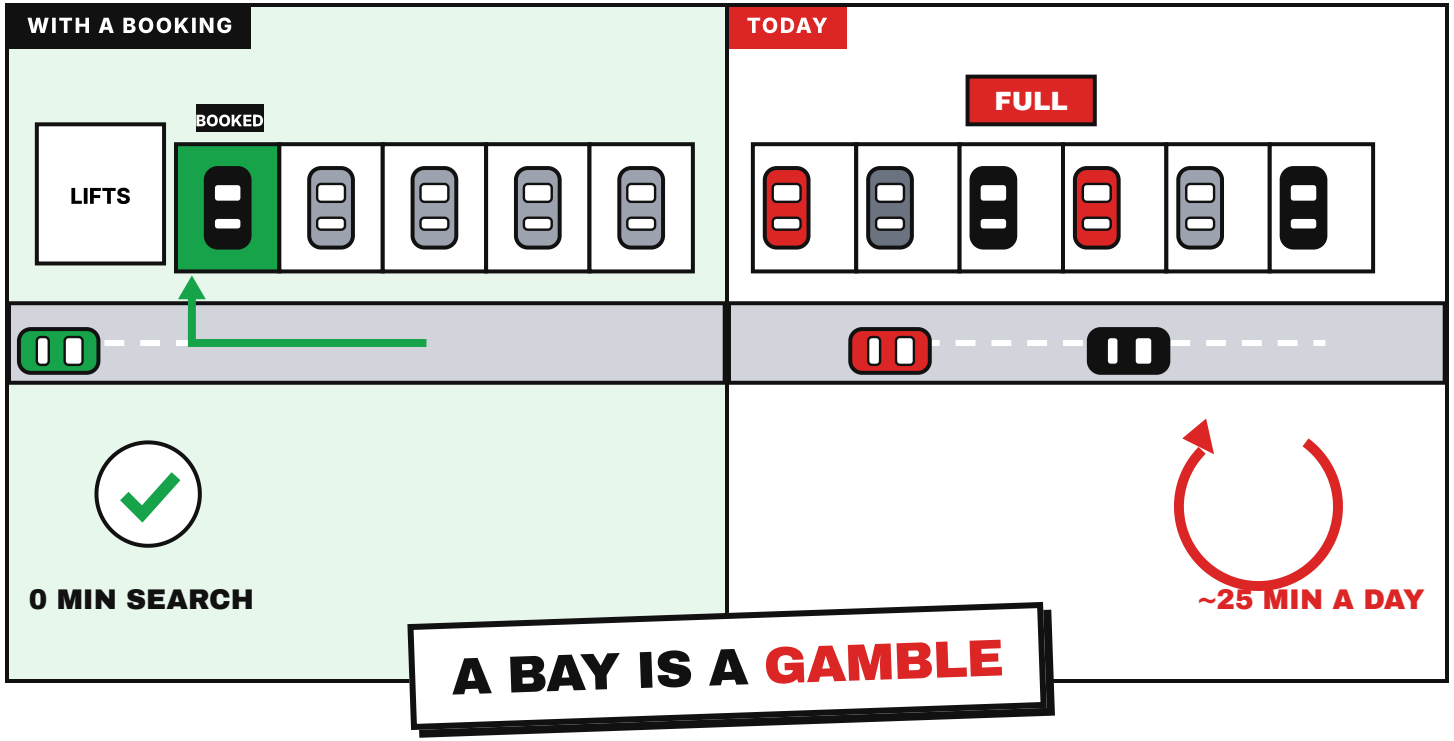
TEAM	LEADER	UNIVERSITY
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ALREADY BUILT Forecast tested on **30 real car parks** · 1,000-day Monte Carlo simulation · costed business case · working prototype

THE PROBLEM · 01

AT PEAK, NOBODY CAN COUNT ON A BAY.
MALLS LOSE THE VISIT.



WHO LOSES — **Shoppers** circle · **loyal customers** get no help · **malls** lose visits to easier rivals

THE SEARCH TAX

~25 min

A DAY KL DRIVERS SPEND HUNTING FOR PARKING

Avoided shops	40%
Gave up a trip	26%
Would pay to reserve	67%

WHY SDP CARES

10%

RECURRING INCOME TODAY · GOAL 30%

Today	10%
Goal	30%

Parking is every visitor's first experience at four township malls.

Sources: BCG "Unlocking Cities" 2018 · Müller et al. 2026 · INRIX 2017 · ParkMobile 2024 · YSDAF KL East Mall guide · SDP

TARGET USER · 02

THREE GROUPS LOSE OUT TODAY

CUSTOMER

SDP'S MALLS

Need
A reason to be the mall that wins when parking is tight.

Without
Hardware on every bay, or bays lost to no-shows.

USER

BEST CUSTOMERS

Need
A reward felt every visit and a bay at peak.

Today
The app gives vouchers, not parking.

USER

EVERYONE ELSE

Need
A bay at a set time: film, pram, older parent.

Today
They circle. Reserved bays go unenforced.

START HERE

1,300
KL EAST MALL BAYS · SDP-OWNED

ROLL-OUT

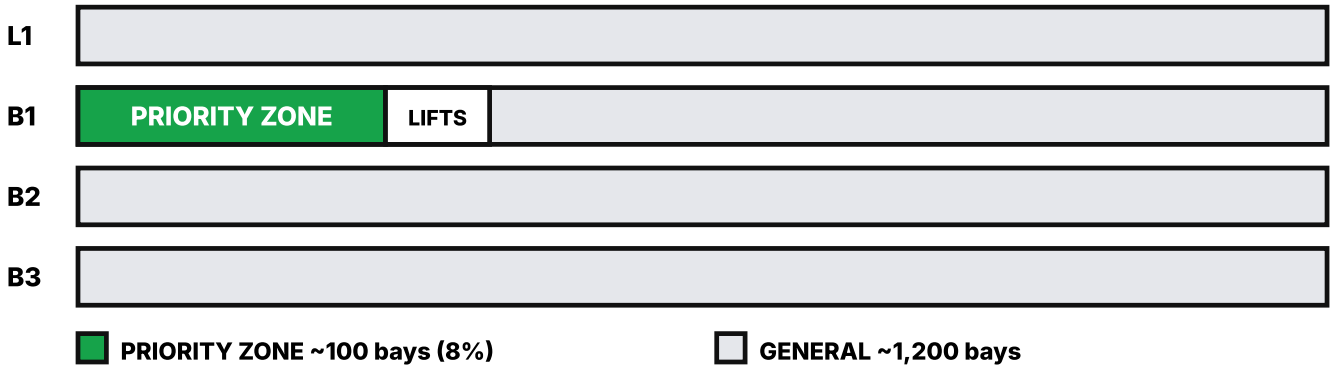
KL EAST → KLGCC ELMINA LAKESIDE MELAWATI → OTHER MALLS WITH REWARDS PROGRAMMES

KL East Mall's app (built for SDP) already issues QR shopping vouchers; ParkFlow adds a booking tab. 1 Utama ONECARD shows parking works as a perk.

YOUR SOLUTION · 03

A BOOKABLE PRIORITY ZONE BY THE LIFTS

PILOT DESIGN · KL EAST MALL · ~1,300 BAYS



BOOKABLE
In the mall app or by QR. Any bay in the zone.

ENFORCED
One plate camera. Non-bookers pay a premium.

SHARED
The forecast sets the sign, green or amber, every minute.

WHO BOOKS

Anyone kept on a no-show	RM4 FEE
Big spenders RM300+ in a day → voucher	FREE
Cinema & dining sponsored by the tenant	FREE
Care booking seniors, OKU, prams	FREE

GENERAL PARKING

92%

OF THE CAR PARK: NOTHING CHANGES

Each level forecast 30, 60, 90 min ahead, in the app and on entrance signs.

Bookings held until 30 min after the booked time. Precedents: Sunway Pyramid bookable bays (2019); 1 Utama ONECARD member parking.

INNOVATION · 04

HOLD ONLY WHAT BOOKED DRIVERS USE. SHARE THE REST.

EXISTING APPROACH	LIMIT
Member parking areas 1 Utama ONECARD	Fixed area, held even when empty
Reserved VIP / family bays Mid Valley, Jaya One	Rarely enforced: **34%** misused (UK)
Bay locks, booking apps SpotHero, JustPark	Fixed share, walk-ins locked out
Learned capacity sharing Xie 2025; Müller 2026	Black box staff can't check
ParkFlow	**Transparent rule, every minute, on existing cameras, tied to rewards**

RESEARCH Airline-style revenue management. Mixing reserved and walk-in parking raised use **21%** in simulation.

01

CAPACITY, NOT BAYS

No hardware per bay. One plate camera and one sign per zone.

02

PARKING IS A REWARD

Free booking vouchers for big spenders, inside the app they already use.

03

FOUR PAYERS, ONE ZONE

Drivers, tenants, the mall's rewards budget, licensed malls.

1 Utama: SAYS (2025). Misuse: Confused.com / PCP 2022. Teodorović & Lučić 2006; Kotb et al. 2016.

HOW IT WORKS · 05

ONE FORECAST, TWO DECISIONS

A Priority Zone: green or amber?

6:30 pm · 8 bays free · 15 booked cars due by 7:00

COUNTING ONLY

8 − 15 < 0

Not enough room.

AMBER · WALK-INS TURNED AWAY

WITH FORECAST

8 + 13 − 14 = 7

13 cars leaving, 14 booked will actually come.

GREEN · WALK-INS WELCOME

BUSY SATURDAY

Bays held back while walk-ins wait: **12 → 5** bay-hours. Every booking safe. Forecast fails? Sign falls back to counting.

B General parking: which level fills when?

PREDICTS

Each level, 30 / 60 / 90 min ahead, with an honest 80% range.

INPUTS

Gate and camera counts, GSC showtimes, bookings, weather.

OUTPUTS

App, entrance signs, early warning for the mall team.

TESTED ON 30 REAL CAR PARKS

61%

MORE ACCURATE THAN "SAME TIME LAST WEEK"



TWO BLENDED MODELS

A **pattern model** and **LightGBM** machine learning, blended. Staff can audit the rule behind every sign.

Counting cannot know how long each car will stay, film end times or past show-up rates; the forecast can. Evidence: appendix A1, A6.

IMPACT · 06

EVERY BOOKING KEPT. A QUARTER OF THE BAYS HELD.

BOOKED DRIVERS

0 MIN

searching, every time

EVERY DRIVER

−69%

search time with level guidance

SDP'S MALLS

95%

zone used at peak vs 91% for a VIP section

PLANET

1.2 T

CO_2 saved per busy Saturday

BAYS HELD WHILE WALK-INS WAIT

5

BAY-HOURS PER BUSY SATURDAY



BOOKINGS THAT GOT A BAY

100%

OF 1,000 SIMULATED BUSY SATURDAYS



WORST CASE

Forecast wrong on purpose, 10% ignore the sign: ParkFlow keeps **99.6%** of bookings; a VIP section 93%.

Monte Carlo: 1,000 busy Saturdays, 1,300 bays. Research inputs: Müller 2026, Sun 2025; show-up 80% free / 95% paid. Our assumptions: demand, 60 bookings a day. Limits: A1; evidence: A15.

PROVEN PARTS. PAYBACK IN 2.4 YEARS.

TECHNOLOGY

- Plate-reading entry is live in Malaysian malls
- Camera bay detection: 97–99%
- New part: the zone rule

RESOURCES

- 1 plate camera, 1 LED sign
- Booking tab in the existing app
- ~**RM43k** hardware per mall

PARTNERS

- **Mall:** zone, signs, vouchers
- **Operator:** gate data
- **SDP IT / DPO:** app, PDPA

FOUR PAYERS, ONE PLATFORM

Drivers	**RM4 / booking**
Tenants	**RM2 / sponsored bay**
Mall rewards budget	**Free bookings**
Licensed malls	**RM2,500 / zone / month**

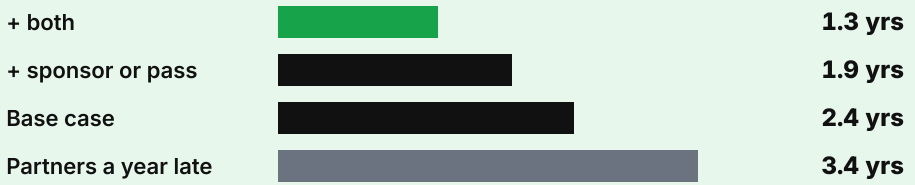
MARKET

173 Klang Valley malls · **490** in Malaysia. Sold to mall groups: 6 partner malls a year from year 2, priced per zone. SOM in year 3: **RM0.66m**.

PAYBACK

2.4 yrs

FEES, TENANT SPONSORS AND PARTNER LICENCES



Card sponsor: a bank pays RM2,000 a month per mall to name the zone. **Priority Pass:** 150 shoppers per mall pay RM20 a month. Both are tested in the pilot (A4).

DEPLOYMENT

One zone first, run by our team for **8 weeks**; then SDP digital or a licensed partner. SDP's edge: first mover across four malls, with measured results.

Fees: 20–40 paid bookings × RM4 on 104 weekend days, 5–10 on others, after the operator's 20%. Costs: list prices + our estimates, quotes to follow (A4, A14). Payback model: A4. Risks: A11.

IMPLEMENTATION · 08

8-WEEK PILOT AT KL EAST MALL

1

PROTOTYPE

Built: booking tab, zone sign, dashboard on simulated feed

→

2

PILOT · 8 WEEKS

100-bay zone by GSC lifts. Wk 1–2 measure; wk 3–8 alternate fixed vs forecast

→

3

MEASURE

Keep the forecast only if it beats the fixed rule

→

4

SCALE

3 more SDP malls → member tiers & points → partners

GO / NO-GO AT WEEK 8

99%+
Booked drivers who got a bay

30+
Bookings per busy day

−30%
Search time vs weeks 1–2

95%+
Car-count accuracy

≤3%
Forecast error, 60 min ahead

100%
Care bookings honoured

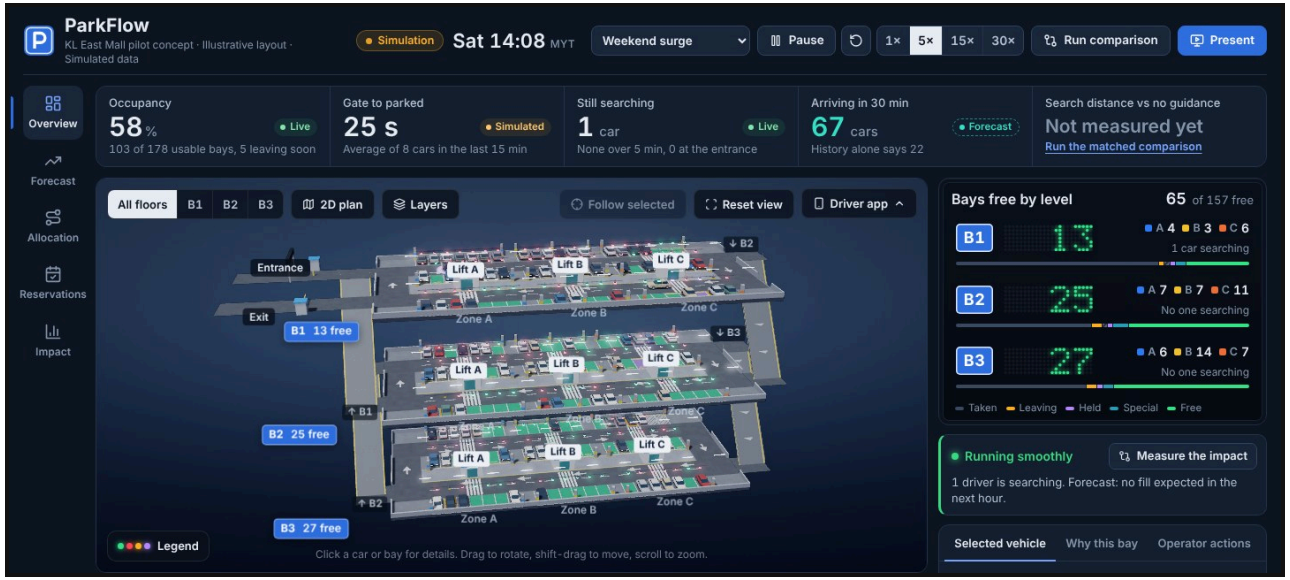
Also measured: bays held while walk-ins wait (fewer than fixed-rule days), vouchers redeemed, spend per visit, CO_2 saved.

WEEK 8 **SDP chooses: scale · adjust · stop.** 10 busy days are enough to tell.

What we validate before the final: appendix A14.

VISUAL CONCEPT · 09

THE WORKING PROTOTYPE, PLUS THE BOOKING WE ADD NEXT



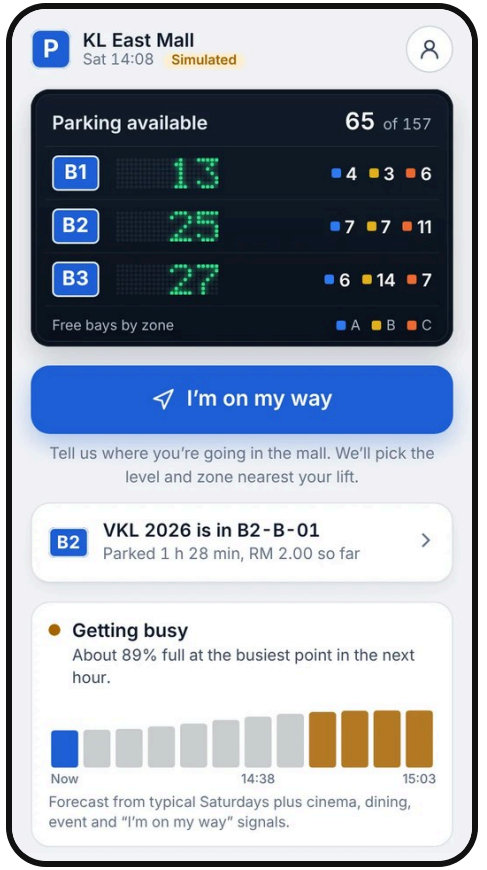
REAL PROTOTYPE Command centre: live 3D car park, free bays by level, 30-minute arrivals forecast

ZONE OPEN TO ALL
8 free · bookings covered

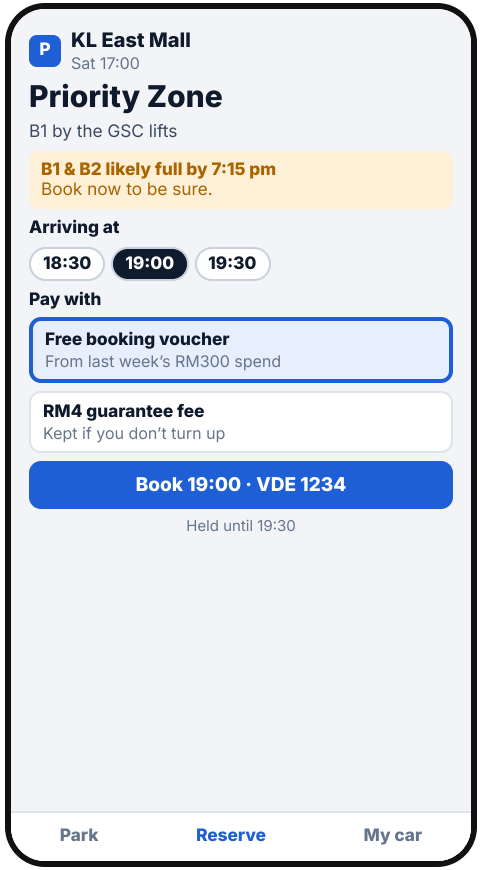
BOOKED CARS ONLY
B3 has 40 free

ZONE FULL
B3 has 40 free

DESIGN Entrance sign. The forecast sets it every minute.



REAL PROTOTYPE Driver app: free bays, "I'm on my way", "Getting busy"



DESIGN MOCK-UP Priority Zone booking, next

Prototype screens use simulated data on an illustrative 157-bay layout (command.html, driver.html in our submission).

BUILT. TESTED. READY TO PILOT AT KL EAST.

BOOKINGS KEPT
99.6–100%
over 1,000 Monte Carlo Saturdays

BAYS HELD BACK
5 VS 19
bay-hours: ParkFlow vs a VIP section

FORECAST
61%
more accurate, 30 real car parks

PAYBACK
2.4 YRS
1.9 with a card sponsor or Priority Pass

JUDGING CRITERION	OUR PROOF
Impact	Booked shoppers search 0 min; level guidance cuts everyone's search ~69%; free care bays
Feasibility	Existing cameras, KL East Mall's own app, ~RM43k hardware, 8-week pilot with go/no-go targets
Innovation	Not a booking app with AI added: a zone shared by forecast, and parking turned into a reward
Commercial	Sold to mall groups, priced per zone: SOM RM0.66m in year 3; payback 2.4 years
Fit & team	**SDP-owned mall and app; working prototype; every number reproducible from our code**

WHAT WE BUILD NEXT

EV CHARGING BAYS
EV registrations doubled in 2025 (**+106%**) while public chargers grew ~43%. A bookable charging bay is next.

PAY BY PLATE
300,000+ drivers pay for parking in Touch 'n Go every day. Booking, entry and payment in one flow.

TIERS & POINTS
A **40-year** meta-analysis finds loyalty programmes reliably lift repeat visits. Parking is felt every visit.

Judging weights (briefing): impact 25%, feasibility 20%, innovation 20%, commercial 10%. EV: JPJ via NST (Nov 2025); chargers: SoyaCincau (Oct 2025); Touch 'n Go Digital (2023); Belli et al. 2022.

THANK YOU

RESERVE CAPACITY, NOT PARKING BAYS.

OUR ASK: AN 8-WEEK PILOT ON ONE ZONE OF KL EAST MALL

Team Hexa · Universiti Malaya · Leow Shen En (leader), Pohyi Chong, Isaac Teo, Lee Jia Shern, Choong Zhuo Lin
leowshenen@gmail.com

Questions? The appendix answers likely questions: next slide, or press A.

APPENDIX

ANSWERS TO LIKELY QUESTIONS

THE EVIDENCE AND THE BUSINESS

A1 What does our evidence prove, even if the forecast is wrong?

A2 How much search time and CO2 does it save?

A3 How big is the market, and who pays?

A4 What does it cost, and when does it break even?

A5 How big should the Priority Zone be?

A6 How accurate is the forecast?

A7 How does the forecast engine work?

A8 Will the parking operator agree?

RUNNING IT

A9 Booking rules: who pays, and who books free?

A10 Privacy: counting cars without tracking them

A11 What could go wrong?

A12 How does it fit a mall's rewards programme?

A13 Barrier or no barrier at the zone?

A14 What we validate before the final, and when

A15 Which numbers come from research?

A16 Why not buy from a vendor? Couldn't others copy it?

Click a question to jump to it. Press A to come back here, Home to return to the start.

APPENDIX

A1. BOOKINGS STAY SAFE WITH A LEARNED MODEL AND DAY-TO-DAY SHOCKS; IF MANY DRIVERS IGNORE AMBER, THE ZONE SWITCHES TO THE SIMPLE RULE

BUSY SATURDAY, 1,000 MONTE CARLO DAYS: BOOKINGS THAT GOT A BAY	VIP SECTION	FIXED RULE, NO SAFETY BAYS	PARKFLOW
As assumed	99.94%	100%	100%
Stays 20% longer, more show-ups, 10% ignore amber	93.4%	99.9%	99.6%
Stay model learned from a 2-week pilot, 40% cinema-goers	99.9%	100%	100%
Day-level shocks: all stays and show-ups shift together (sd 30%)	99.2%	100%	99.99%
20% of shoppers ignore amber	94.7%	99.93%	99.4%
30% ignore amber	**92.6%**	**99.7%**	**98.5%**
30% ignore + shocks + learned model + cinema-goers	87.5%	97.8%	95.7%
Bays held back while walk-ins waited (bay-h, as assumed)	**19**	**12**	**5**
Bays held back in all minutes (bay-h, as assumed)	143	62	33

WHAT IT PROVES

With a model learned from two weeks of data, mis-shaped stays and day-level shocks, ParkFlow keeps 99.99–100% of bookings while holding back a quarter of the bays a VIP section does while walk-ins wait (5 vs 19 bay-hours).

HONEST LIMIT, AND THE FIX

When 20% or more ignore amber, the simple rule keeps more bookings (99.7% vs 98.5% at 30%). The plate camera counts amber violations live: above 10%, the zone runs the simple rule until they fall.

Only the pilot can prove: real stays and show-ups at KL East Mall, how many people book, and how shoppers react to an amber sign. Demand levels in these runs are our assumption (A15).

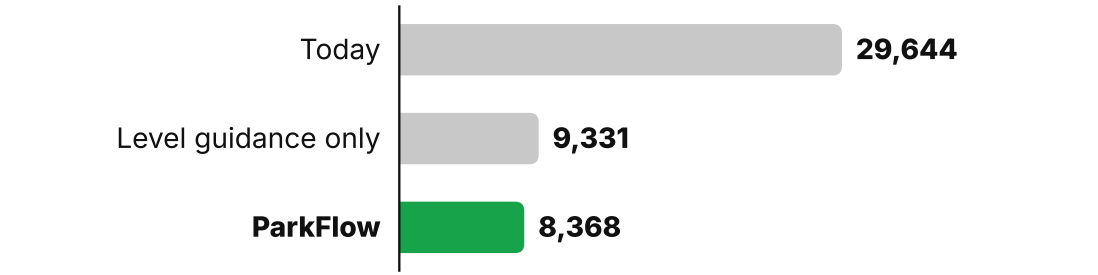
Monte Carlo: each day draws arrivals, stays and show-ups at random from research-based distributions. Learned model: 50 simulated 2-week pilots, each fitting the stay model the rule then uses; show-up rates stay at published priors. Shocks: one random factor a day for all stays and show-ups that the rule cannot see. Script: analysis/simulate_pooling.py.

APPENDIX

A2. WHERE A MALL HAS NO LEVEL GUIDANCE, ADDING IT CUTS SEARCH TIME BY ABOUT 69%; THE ZONE ADDS ZERO-MINUTE SEARCHES FOR BOOKED DRIVERS

Minutes all drivers spend searching

Busy Saturday, 1,300 bays, 1,000 Monte Carlo days



BUSY SATURDAY	TODAY	GUIDANCE ONLY	PARKFLOW
Search minutes per driver	5.7	1.8	1.6
Drivers searching over 10 minutes	777	0	0
Walk-ins' total search minutes	29,226	9,271	8,368
Booked drivers' search minutes	419	60	0
Saved by level guidance (min / CO2)	–	20,313 / 1.17 t	20,313 / 1.17 t
Extra saved by the zone (min / CO2)	**–**	**–**	**963 / 0.06 t**

HONEST ATTRIBUTION

About 95% of the saving is level guidance, which count-based systems also give. ParkFlow's own share is the zone: booked drivers search 0 minutes. If KL East has guidance already, only the zone's share counts (site visit, A14).

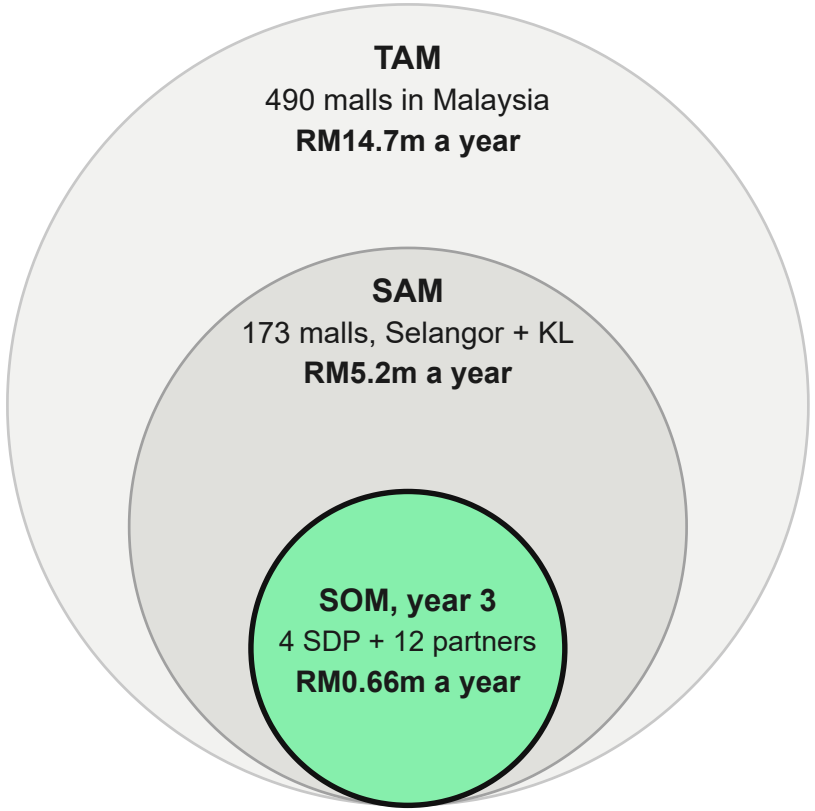
WHAT WE WOULD REPORT

Guidance and zone separately, measured in the pilot, not modelled. Upper bound per mall: ~60,000 driver-hours and ~200 t CO2 a year if every weekend is this busy and there is no guidance today.

Monte Carlo model: every car in 4 levels of 325 bays; today drivers try B1, B2, B3, L1 and learn a level is full by circling it. The zone uses the simple fixed rule (conservative). Cruising 10 km/h, 0.15 L/km, 2.31 kg CO2/L petrol (DESNZ). Script: analysis/simulate_carpark.py.

APPENDIX

A3. MARKET AND BUSINESS MODEL: SELL TO MALL GROUPS FROM YEAR 2, PRICED PER ZONE, FOR A YEAR-3 SOM OF RM0.66M



WHO PAYS, AND FOR WHAT

PAYER	WHAT THEY GET	PRICE
Drivers	A guaranteed bay in the Priority Zone	RM4 a booking
Tenants: cinema, restaurants	Sponsor a bay for their customers ("film + guaranteed bay")	RM2 a booking, tested in the pilot
The mall's rewards budget	**Free booking vouchers for big spenders now; free bookings for top-tier members in phase 2**	**Funded as loyalty**
Mall groups	Licence per zone for each mall, linked to their own rewards programme; they keep booking fees	RM2,500 a month per zone
Card issuer (pilot target)	Its name on the zone; its cardholders book free	RM2,000 a month per mall
Regular shoppers (pilot target)	Priority Pass: unlimited Priority Zone bookings	RM20 a month

VALUE BEYOND FEES (NOT COUNTED IN A4)

- **Rewards:** a perk that gives shoppers a reason to join and to stay in the top tier
- **Tenants:** customers sure of a bay; spend per visit and turnover rent measured in the pilot
- **Planet:** shorter searches for every driver (A2)

TAM and SAM: number of malls × RM30,000 a year (licence value). Malls: PPK Malaysia, 490 at end-2024 (The Star); NAPIC, Selangor 95 + KL 78 enclosed malls, mid-2025 (The Rakyat Post). SOM: 4 SDP malls + 12 partner malls in year 3, sold to mall groups, partners averaging 1.5 zones (larger malls run 2–3). Script: analysis/business_case.py.

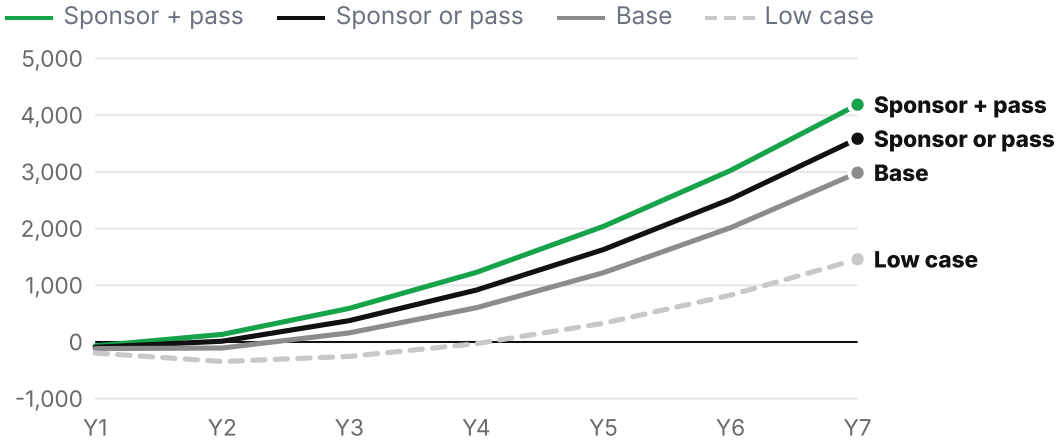
APPENDIX

A4. PAYBACK IN 2.4 YEARS WITH MALL GROUPS FROM YEAR 2; 1.9 WITH A CARD SPONSOR OR PRIORITY PASS; 3.4 IF PARTNERS COME A YEAR LATE

COST AND REVENUE	BASE	RANGE
Hardware per mall, one-off: cameras, sign, AI boxes, install	**RM43,140**	**30–58k**
Platform build, once, on our prototype	RM60,000	45–100k
Running per zone a year: cloud, WhatsApp, upkeep	RM10,500	7.9–14.9k
Engineering support a year (0.4 of an engineer)	RM48,000	36–72k; doubles past 12 malls
Partner onboarding and support a year	RM30,000	From the first partner
Revenue per SDP mall a year: fees after the operator's 20%, plus tenant sponsors	**RM21,050**	**12.6–29.5k**
Licence per partner mall a year: RM30,000 a zone × 1.5 zones	**RM45,000**	**6 malls a year from year 2, via mall groups**
Card sponsor per SDP mall a year (pilot target)	RM24,000	RM2,000 a month
Priority Pass per SDP mall a year, net (pilot target)	RM24,048	150 × RM20 a month, less the RM4 bookings it replaces

Cumulative cash by scenario, RM thousands

KL East in year 1; 3 more SDP malls and 6 partner malls in year 2; 6 partners a year after



SCENARIO	PAYBACK
Base: fees, tenant sponsors, partner licences	**2.4 years**
+ card sponsor, or + Priority Pass	1.9 years
+ both	1.3 years
Partners from year 3 instead of year 2	3.4 years
Low case: fewest bookings, highest costs	4.1 years

SDP's 4 malls alone, no partners: a net cost of RM52k a year with build and hardware spread over 5 years (about RM13k per mall). With a card sponsor and Priority Pass they earn RM140k a year and pay back in 2.1 years.

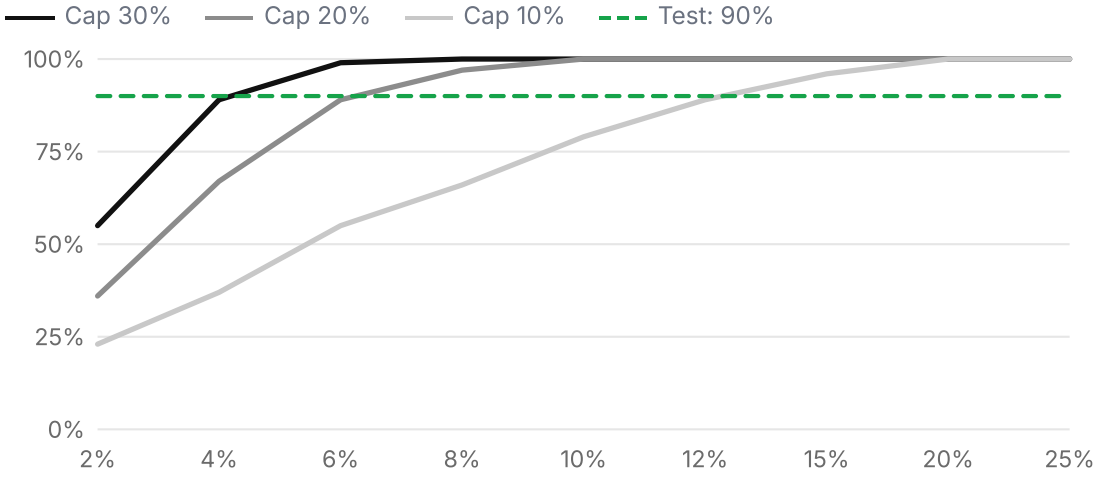
Partner malls buy their own hardware. Platform build and engineering are sized for our prototype and a small portfolio; hardware uses list prices and estimates until quotes arrive. The premium rate from non-bookers goes to the operator (A8) and is not counted; nor are rewards sign-ups, spend or turnover rent. Script: analysis/business_case.py.

APPENDIX

A5. HOW BIG SHOULD THE PRIORITY ZONE BE? ABOUT 6–10% OF THE CAR PARK, WITH 20–30% OF IT BOOKABLE AT ONCE

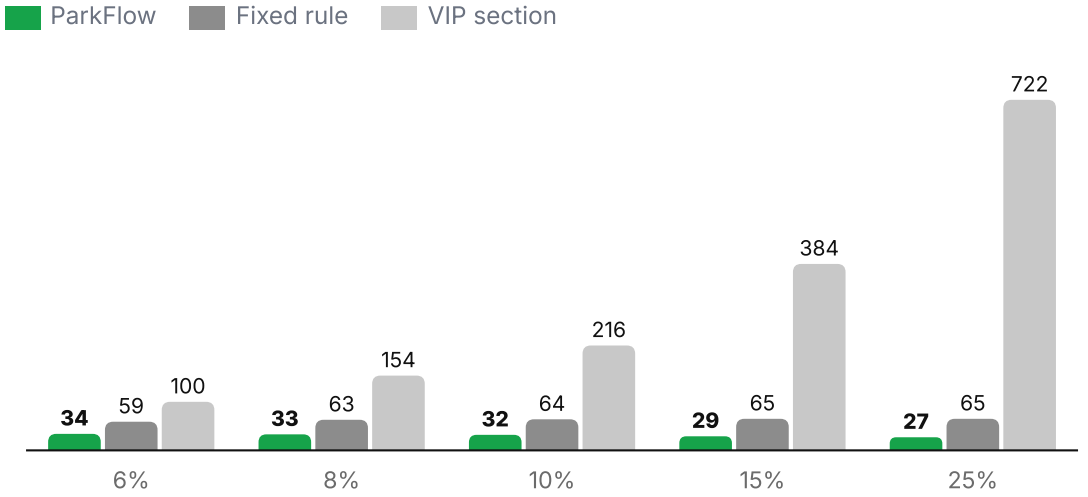
Booking requests accepted

Busy day, by zone size (% of car park) and booking cap (% of zone)



Bays held back from walk-ins (all minutes)

Busy day, bay-hours, by zone size (% of car park), cap 20% of the zone



RECOMMENDED

6–10% of the car park (78–130 bays in a 1,300-bay mall), bookings capped at **20–30% of the zone**. Our 100-bay pilot sits inside this band.

SMALLER OR BIGGER?

Below 6%, the zone turns away 11–77% of busy-day requests, though every booking it takes still gets a bay. Above ~12%, little improves.

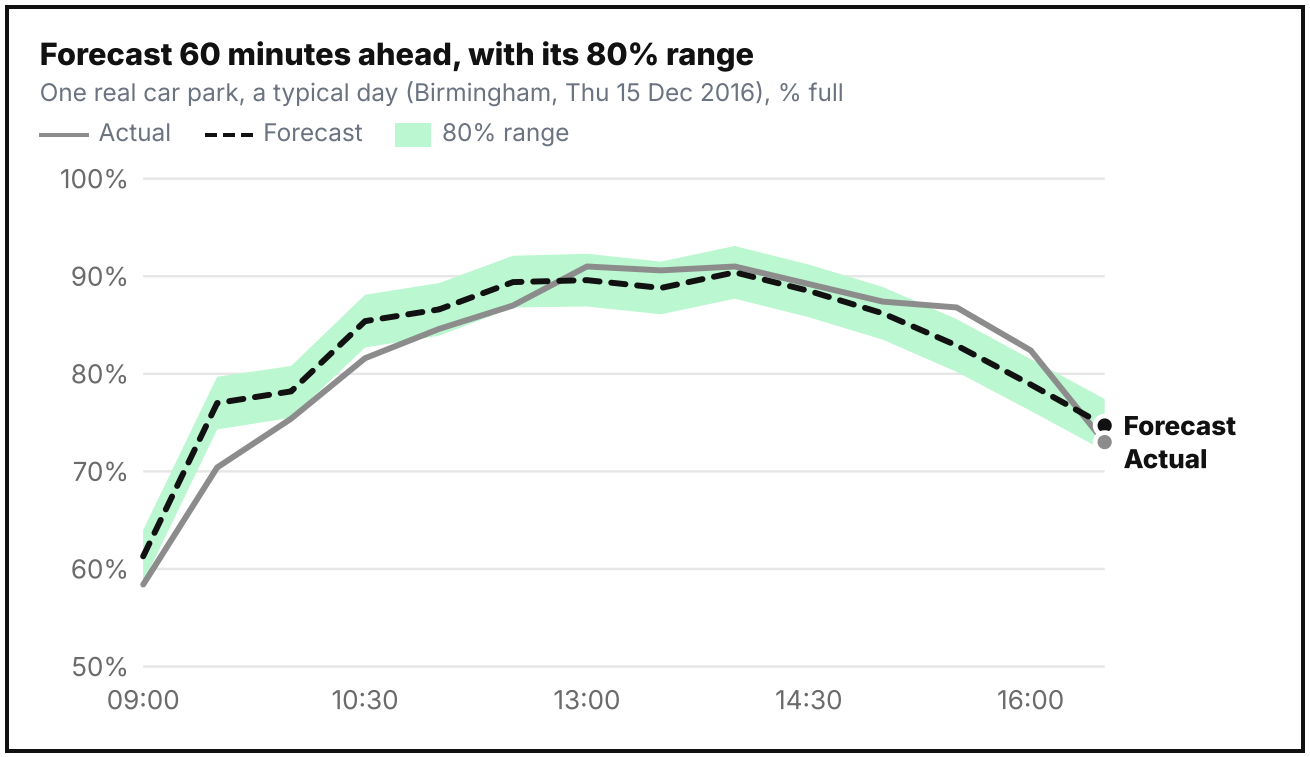
ONE BLOCK OR ACROSS LEVELS

Pieces on several levels run as one pool work like one block. Run as 4 separate pieces, bays held empty rise ~60%.

1,000 Monte Carlo days per setting: 9 zone sizes × 3 caps × busy, normal and stress days. Choice rule set before running: the smallest zone that keeps ≥99.5% of bookings on a stress day and accepts ≥90% of busy-day requests. Script: analysis/simulate_zone_size.py.

APPENDIX

A6. TESTED ON 30 REAL CAR PARKS: OUR BLEND IS 61% MORE ACCURATE THAN "SAME TIME LAST WEEK"



AVERAGE ERROR, % OF CAPACITY (LOWER IS BETTER)

MODEL	30 MIN	60 MIN	90 MIN
Same time last week	5.3	5.3	5.4
Pattern model	1.5	2.2	2.8
LightGBM	1.6	2.2	2.8
Blend (ours)	**1.4**	**2.1**	**2.6**

The blend beats either model alone by 5–10%. Most of the gain over "last week" comes from the pattern model itself.

Honest ranges: the 80% range held the real value 79.8–79.9% of the time. That lets the zone rule be cautious by the right amount.

Data: UCI Parking Birmingham, 30 car parks, 48,112 forecasts over 6 weeks, each week predicted only from earlier weeks; daytime only, no weather or event inputs. The day shown is typical (median error). Departures and show-ups use the same engine and are calibrated on KL East Mall's data in pilot weeks 1–2. Script: analysis/backtest_ensemble.py.

APPENDIX

A7. FORECAST ENGINE: TWO MODELS, BLENDED, WITH AN HONEST RANGE

INPUTS

- Gate and plate-camera times
- CCTV bay counts
- Bookings × show-up history
- Calendar, GSC shows, weather

TWO MODELS

Pattern model
What this hour usually looks like, adjusted to today

LightGBM
Gradient-boosted trees: how events, weather and bookings change it

BLEND

Stacked ensemble
Weights re-learned each week from each model's recent errors

RANGE

Conformal prediction
Past errors become an honest 80% range

OUTPUT

Zone rule: cautious departures and show-ups

Mall team: fan chart and top reasons



Industry standard. Combining models is the main lesson of the M4 forecasting competition: 12 of the 17 most accurate methods were combinations. LightGBM was used by most top teams in M5. Conformal prediction is the standard way to make forecast ranges honest.



RUNS ITSELF, FAILS SAFE

Retrains every night and tracks its own error. If a data feed breaks or bookings start missing bays, the zone falls back to the fixed rule.

Roadmap. Pilot: this ensemble. After 6–12 months of mall data: test a Temporal Fusion Transformer as a third member, kept only if it beats the ensemble. Later: one shared model across SDP's four malls.

APPENDIX

A8. WILL THE PARKING OPERATOR AGREE? YES, IF THEY GAIN TOO, AND SDP HAS A FALLBACK

WHAT WE NEED FROM THE OPERATOR	WHAT THE OPERATOR GETS
An anonymised feed of entry and exit times (data export or API)	Fewer queues and complaints at their gates
A discount at exit for approved plates (fee refunds, points payments)	A share of booking fees
The posted premium rate at exit for non-bookers in a held zone	The premium-rate income, and peak forecasts to plan staff

SDP'S LEVER

As landlord, SDP can make data sharing part of the operator contract at the next review.

FALLBACK

Our own zone plate camera and floor-entrance cameras give counts and stay times without the operator's feed. The exit discount becomes a voucher or points.

To confirm in the mentoring sprint: which operator runs KL East Mall's car park, which system it uses, and when its contract is reviewed.

APPENDIX

A9. BOOKING RULES: FREE FOR BIG SPENDERS AND FOR THOSE WHO NEED IT MOST, RM4 FOR ANYONE ELSE, AND RULES THAT STOP "JUST IN CASE" BOOKINGS

RULE	ANYONE	VOUCHER (BIG SPENDER OR TENANT-SPONSORED)	PHASE 2: GOLD MEMBER
Price	RM4 guarantee fee	Free	Free
Care booking	**Free for anyone: seniors 60+, OKU companions, prams, pregnancy. By app, phone or at the concierge**		
Book ahead	Up to 2 hours, or by QR at the zone	Up to 24 hours	Up to 24 hours
Bookings at once	One per plate	One per voucher	One per account
Cancel	Refund up to 30 min before	Voucher back up to 30 min before	Free up to 30 min before
Held until	30 minutes after the booked time (15-minute window + 15 minutes grace), then released		
No-show	Fee kept	Voucher used up	Counts against free bookings



WHY CHARGE AT ALL?

Free bookings get made "just in case": 9–13% of free restaurant bookings are no-shows (Tse & Poon 2016). So we expect ~80% of free and ~95% of paid bookings to turn up.



WHY VOUCHERS

KL East Mall's app already runs QR vouchers. A no-show uses the voucher up, which discourages "just in case" bookings. We still model vouchers at the cautious free show-up rate (80%).

No booking: green = park as usual · amber = go to the level shown, book by QR if there's room, or park and pay the premium rate · red = full. A car that entered on green pays the normal rate even if the sign turns amber later. OKU bays stay outside the zone.

RM300 threshold and the Gold tier are examples for the mall to set. Precedents: Sunway Pyramid bookable bays RM6–8, held 1 hour (Autobuzz 2019); ParkEasy KL ~RM3–4 (Vulcan Post 2015).

APPENDIX

A10. PRIVACY: WE COUNT CARS, NOT PEOPLE, AND KEEP LITTLE PERSONAL DATA, BRIEFLY, ON SDP'S TERMS

DATA	WHERE IT LIVES	HOW LONG	WHO SEES IT
Camera video	Processed on the AI box in the mall, never sent out or stored by us	Not kept	No one
Bay and zone counts	Cloud database	Kept, to improve forecasts	Mall team
Plates of non-bookers	Scrambled with a secret key that changes daily	Deleted after 24 hours	No one, only counts and stay times
Plates of bookers	Linked by the driver's own consent in the KL East Mall app or QR booking	While the account is active; 30 days for one-off bookings	The driver; the system for bookings and points



COUNTING, NOT TRACKING

Bay cameras see whether a bay has a car, never whose. Plates are read only at the gate and zone entrance. The AI box sits on its own network segment.



SIGN-OFF

SDP's data protection officer and IT security team review it before launch. PDPA notices go up at every entrance.



LAW

Built for the PDPA as amended in 2024, including the duty to report data breaches.

APPENDIX

A11. THE MAIN RISKS ARE ACCESS AND ADOPTION, NOT TECHNOLOGY, AND EACH HAS A PLAN

RISK	HOW WE HANDLE IT
Drivers ignore amber, or a booked driver finds the zone full	The camera counts amber violations live; above 10% the zone runs the simple rule. A booked driver without a bay gets the nearest free bay, a refund and free parking; the concierge is alerted.
Free (voucher) bookings made "just in case"	One booking per voucher; a no-show uses it up; release 30 minutes after the booked time. The forecast learns show-up rates.
KL East's app has vouchers but no member tiers	Launch with vouchers in the existing app; add tiers and points in phase 2. Partner malls plug in their own programme.
Non-bookers feel shut out	Amber only when needed; the sign always says where free bays are; anyone can book for RM4. OKU bays are never held.
Mall groups sign a year late	Partners from year 3 still pay back in 3.4 years. A card sponsor or Priority Pass at SDP's malls makes up most of the gap (A4).
No card sponsor signs	Not needed for the base case (2.4 years); a Priority Pass alone brings it to 1.9 years.
Parking system can't discount by plate	Give a voucher or points for the next visit instead.
Parking operator is slow to share gate data	SDP asks for it under the operator contract. Fallback: our own zone and floor-entrance cameras.
Privacy concerns	Cameras never identify cars. Non-member plates scrambled daily and deleted after 24 hours. PDPA notices at entrances.

APPENDIX

A12. HOW PARKFLOW FITS A MALL'S REWARDS: VOUCHERS IN KL EAST MALL'S APP TODAY, MEMBER TIERS AND POINTS NEXT

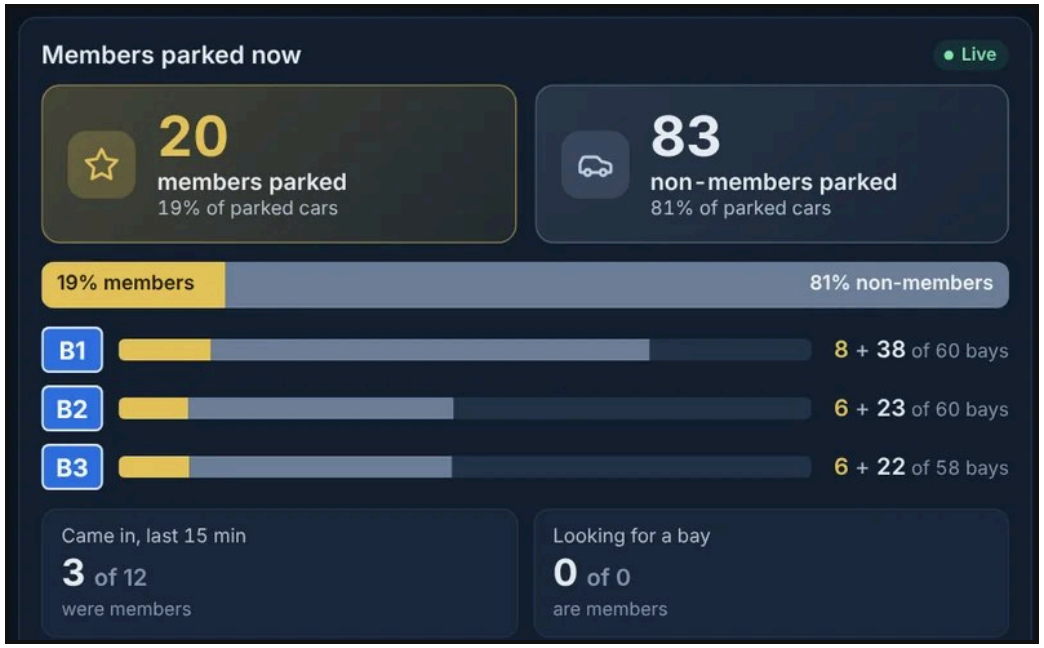
HOW MALL REWARDS TREAT PARKING TODAY

PROGRAMME	REWARDS TODAY	WHAT PARKFLOW ADDS
KL East Mall app (SDP)	**QR shopping vouchers for everyone; no parking perk found**	**Free Priority Zone booking vouchers, using the same QR system**
1 Utama ONECARD	Preferred and Express Parking areas; lower member rates (reported RM2 vs RM4 for 4 hours)	A bay guaranteed at a set time, shared when not needed
CLUB IOI (IOI malls)	Cardless points and privileges; no parking perk found	Parking as a perk, paid with points

PHASE 2: EXAMPLE TIERS

TIER	HOW TO GET IT	PARKING PERK
Member	Free to join in the app	Book with 400 points; pay parking with points
Gold	**Top tier, e.g. by yearly spend**	**Free bookings, up to 24 hours ahead**

WHAT THE MALL LEARNS (REAL PROTOTYPE SCREEN)



With consent: how often regular shoppers visit, how long they stay, and spend per visit, for marketing, tenant planning and pricing.

To confirm with SDP: how many shoppers use the KL East Mall app, voucher redemption rates, and plans for member tiers. Points rates shown use the prototype's 10 points per RM1 and 100 points = RM1.

Sources: KL East Mall app listing (APKPure, v1.0.9, Oct 2024); IT Paradise, KL East Mall app case study (QR voucher redemption); SAYS (Oct 2025), 1 Utama ONECARD parking; IOI Properties, IOI Mall Puchong (CLUB IOI).

APPENDIX

A13. BARRIER OR NO BARRIER? A PLATE CAMERA AND A SIGN HOLD THE ZONE WITHOUT A QUEUE INSIDE THE CAR PARK

OPTION	HOW IT WORKS	STRENGTH	WEAKNESS	VERDICT
Plate camera and sign, no barrier	**Logs every plate entering the zone. Non-bookers who park there while the sign is amber pay a posted premium rate at exit.**	**No queue inside, no change to fire exits, low cost**	**Some ignore the sign: 10% still leaves 99.9% of bookings with a bay; above 10%, the simple rule takes over (A1)**	**Main**
Barrier at zone entrance	Opens only for booked plates while the zone is held	Strict	Queue inside the car park; changes exit routes, so needs Bomba approval	Later, if needed
UHF RFID windscreen tag	Free sticker for regular bookers (EPC Gen2). Backup when a plate can't be read.	Malaysians know RFID from tolls	Metallic window tint can block it	Test first
Lock or bollard on each bay	App unlocks a single bay	Very precise	Costly, upkeep per bay, still empty until arrival	No



ZONE LAYOUT

Placed behind one aisle entry. A car still inside after 10 minutes is parked; one that leaves sooner drove through.



FAIR WARNING AND DISPUTES

The premium rate is posted before entry. Disputes go to the concierge with the camera log; first-time and care-eligible drivers are waived.



FALLBACK

If the plate camera fails, the zone opens to all. If the operator can't bill by plate, the discount becomes a voucher.

APPENDIX

A14. WHAT WE VALIDATE BEFORE THE FINAL: THE SITE, THE SHOPPERS, THE APP, THE OPERATOR AND THE PRICES

WHEN	WHAT WE DO
24–28 Oct	Zone rule engine on a simulated feed; booking tab with the voucher flow, styled for the KL East Mall app
31 Oct – 1 Nov	KL East visit, Sat 5–9 pm: count the lift-side zone and each level, time searches, check for existing level guidance; survey 100 shoppers (do they use the mall app, would they pay RM4, would a free booking voucher bring them back)
2–6 Nov	**Mentoring sprint: meet mall management, the KL East Mall app team (voucher system, app usage), the operator, two card issuers (zone sponsorship) and two mall groups (partner licences); plate discount and premium-rate check; quotes replace A4 estimates**
7–10 Nov	Connect booking tab, zone sign and dashboard. Three test runs.
11 Nov	Final submission: prototype, demo video, field report
14 Nov	Pitch Day



Already done: forecast tested on 30 real car parks; 1,000-day zone simulation with stress tests and a learned model; zone-size sweep; whole-car-park model; business case. All reproducible from the analysis folder.

APPENDIX

A15. WHICH NUMBERS COME FROM RESEARCH, AND WHICH ARE STILL OURS

INPUT	VALUE WE USE	EVIDENCE	STATUS
How long cars stay	Median 87 min, mean ~125	200,000+ plate-camera stays, 1,300-bay mall car park (Müller et al., Applied Sciences 2026)	Research
When cars arrive	Peaks 12–1 pm, 5–6 pm	166,836 mall parking records (Sun et al., PLOS ONE 2025)	Research
Bookings that turn up	80% free, 95% paid or points	Free bookings: 9–13% no-shows (Tse & Poon 2016); deposits cut no-shows	Research
Booking price	RM4	Sunway Pyramid RM6–8 (2019); ParkEasy ~RM3–4 (2015); fee ≈ hourly rate (Xue et al. 2020)	Precedent
Time spent finding parking	~25 min a day, KL	BCG "Unlocking Cities" 2018, via news reports (original report not public)	Survey
Reserved bays misused	34% parent-and-child	Confused.com/PCP 2022	Audit
Forecast accuracy	61% better than last week	Our backtest, 30 real car parks (UCI). Similar: 10.6% error 30 min ahead (Yang et al., TR-C 2019)	Ours + research
Malls reward members with parking	1 Utama ONECARD	SAYS (2025): Preferred and Express Parking areas for members	Precedent
Partner pace; zones per partner	6 malls a year from year 2; 1.5 zones each	None yet: sold to mall groups. Zone size from our sweep (A5)	Ours
Card sponsor; Priority Pass take-up	RM2,000 a month per mall; 150 × RM20 a month	None yet. Precedent: Sunway malls sell season parking from RM140 a month	Ours
Cars wanting the zone; bookings per day	**Zone full at peak; 60 requests**	**None yet: set so the zone fills at peak, the problem being solved**	**Ours**
KL East Mall app users; voucher redemption	**Not used in the simulation**	**App with QR vouchers exists (listing v1.0.9, Oct 2024); usage to confirm with SDP (A12)**	**Open**

The pilot replaces the remaining assumptions with KL East Mall's own gate and booking data in weeks 1–2.

APPENDIX

A16. WHY NOT BUY FROM A VENDOR, AND COULDN'T OTHERS COPY IT? BUY THE HARDWARE, OWN THE RULE AND THE DATA

OPTION	WHAT IT GIVES	WHAT IT MISSES
Upgrade the vendor's guidance system (e.g. Hikvision)	Accurate free-bay counts, per bay	No booking, no rewards link, no forecast. The vendor holds the data.
A generic pre-booking app	Quick to launch	Sells a fixed share of bays. Another app, not linked to the mall's rewards or gates.
ParkFlow	**Shared zone, forecast rule, rewards perks, inside the mall's own app**	**Needs a small team to run it**



COULDN'T ANOTHER MALL BUILD ITS OWN?

Yes: the parts are commodities, and we don't claim a technology moat. SDP's edge is moving first across four malls: data from many car parks improves the forecast, and measured pilot results a newcomer won't have.



LICENSING IS PART OF THE PLAN

ParkFlow works with any rewards programme, so mall groups that already run one, such as 1 Utama or IOI, can license it per zone. With 6 partner malls a year from year 2 it pays back in 2.4 years (A4).

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